

17. Answer: D

To form a stationary wave between the clamp and the pulley, wavelength $\lambda = \frac{2L}{n}$, where n is a positive integer, L is distance between clamp and pulley. n is 1 in the question.

Using $v = f\lambda$ and $v \propto \sqrt{mg}$ (mg = tension applied in the string),

$v \propto \sqrt{m}$ (since g is constant)

$$f\lambda \propto \sqrt{m}$$

$$\lambda \propto \frac{\sqrt{m}}{f}$$

In the question, n is 1,

$$\frac{2L}{1} = k \frac{\sqrt{m}}{f} = k \frac{\sqrt{400}}{15} = \frac{4}{3}k$$

$$\therefore 2L = \frac{4}{3}k$$

Option A: $\frac{2L}{n} = k \frac{\sqrt{500}}{30}$, n cannot be an integer.

Option B: $\frac{2L}{n} = k \frac{\sqrt{600}}{40}$, n cannot be an integer.

Option C: $\frac{2L}{n} = k \frac{\sqrt{800}}{60}$, n cannot be an integer.

Option D: $\frac{2L}{n} = k \frac{\sqrt{900}}{90}$,

$$\frac{2L}{n} = \frac{1}{3}k$$

$$\Rightarrow n = 4$$